

Carbon footprint & GHG accounting

What this tool produces · *A Baseline Emission Inventory (BEI) documenting the tonnes of CO2-equivalent generated across municipal operations, transport, waste, and local industry.*

G1 · Proof of Impact

G4 · Proof of Sustainability

Dimensions

Environmental

Economic

Regulation

Preparing the workshop · *Carbon footprint & GHG accounting*

G1 · Proof of Impact

Why this tool matters

Making the city administratively accountable for the global atmosphere, translating the daily mechanics of turning on lights and driving cars into a shared metric of planetary responsibility.

Evidence we produce

A Baseline Emission Inventory (BEI) documenting the tonnes of CO₂-equivalent generated across municipal operations, transport, waste, and local industry.

Duration & Material

Duration · 2 to 4 months for data collection and calculation

Material · Local utility billing data, municipal fleet records, standardized GHG emission factor tables (e.g., Covenant of Mayors guidelines)

Reference case

Thousands of local authorities participating in the EU Covenant of Mayors rely on compiling a rigorous Baseline Emission Inventory (BEI) as the mandatory first step of their Sustainable Energy and Climate Action Plans (SECAP). By systematically tracking emissions from buildings, transport, and waste, cities like Barcelona establish clear data-driven targets to cut GHG emissions by 2030, ensuring subsequent interventions are measurable and accountable.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

GHG inventory · scopes 1 · 2 · 3

Duration

2 to 4 months for data collection and calculation

Scope 1

direct emissions
fleet · boilers · processes

Scope 2

electricity & district
heat purchased

Scope 3

indirect chain · upstream + downstream
goods · travel · waste · investments

Scope 3 typically dominates · do not skip it.

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.